MIL-S-14249C(ME)
4 August 1981
SUPERSEDING
MIL-S-14249B(ME)
21 May 1976

MILITARY SPECIFICATION

STRAIGHTEDGES, RADIAL-LINE, SLOTTED-TEMPLATE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers radial-line straightedges with pivot holes for slotted templet studs.

1.2 Classification. The straightedges shall be of the following lengths, as specified (see 6.2):

Size A - 12-inch
Size B - 30-inch

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-S-15769	- Straightedges, Steel (Machinists' and Toolmakers').

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 169 - Aluminum Bronze Plate, Sheet, Strip, and Rolled Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The straightedges shall be as shown in Figures 1 and 2, and as specified herein.

3.2 Material. Material shall be bronze conforming to ASTM B 169, copper alloy UNS No. C60600, Temper H04 (see 6.4).

3.3 Construction. The straightedges shall be constructed as shown in Figures 1 and 2, and as specified herein.

3.3.1 Edge straightness. The beveled working edge shall not deviate more than 0.006 inch from a reference plane.

3.3.2 Surface planeness. All surfaces shall be true planes to within 0.006 inch from a reference plane tangent to the edges.

3.4 Identification marking. The straightedge shall be identified in accordance with MIL-STD-130, stamped or otherwise permanently marked on the top surface.

3.5 Workmanship. All surfaces of the straightedges shall be free of sharp edges, cracks, breaks, nicks, chips, burrs, toolmarks, or any other defect which could affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 0.65 percent defective for major defects and 10.0 percent defective for the minor defect.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 0.65 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The straightedges shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Nonconformance to figures.
103. Workmanship not as specified.

Minor

201. Identification marking not as specified.

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4.4.2 Tests.

4.4.2.1 Edge straightness. Place a 36-inch straightedge conforming to MIL-S-15769 on the surface of the beveled ruling edge. Check the contact or lack of contact of the straightedge with the beveled ruling edge with thickness gages along the entire length of the straightedge. Evidence of a gap of more than 0.006 inch between the surface of the beveled ruling edge and the edge of the straightedge shall constitute failure of this test.

4.4.2.2 Planeness. Check the straightedges for planeness by laying the straightedges flat on a machinist's flat, gage block, or true, straight surface plate. A thickness gage clearance at any point along the length of more than 0.006 inch between the straightedge and surface plate shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

104.	Materials, method, or container not as specified. Each incorrect material, method, or container shall constitute one defect.
105.	Gross weight of boxes exceeds 200 pounds.
106.	Strapping not as specified.
107.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each straightedge shall be preserved in accordance with MIL-P-116, Method IC-3.

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5.1.2 Commercial. Straightedges shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The straightedges, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A, except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. The straightedges shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A and B. Level A and B marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The radial-line straightedges are intended specifically for use in the preparation of templets for triangulation by the slotted templet method.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of straightedge required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When strapping shall be other than as specified (see 5.2.1).

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6.3 Classification changes. Changes in classification of the straightedges between this revision of the specification and the previous edition are as follows:

Old Designation	New Designation
Size - 12-inch	Size A - 12-inch
Size - 30-inch	Size B - 30-inch

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

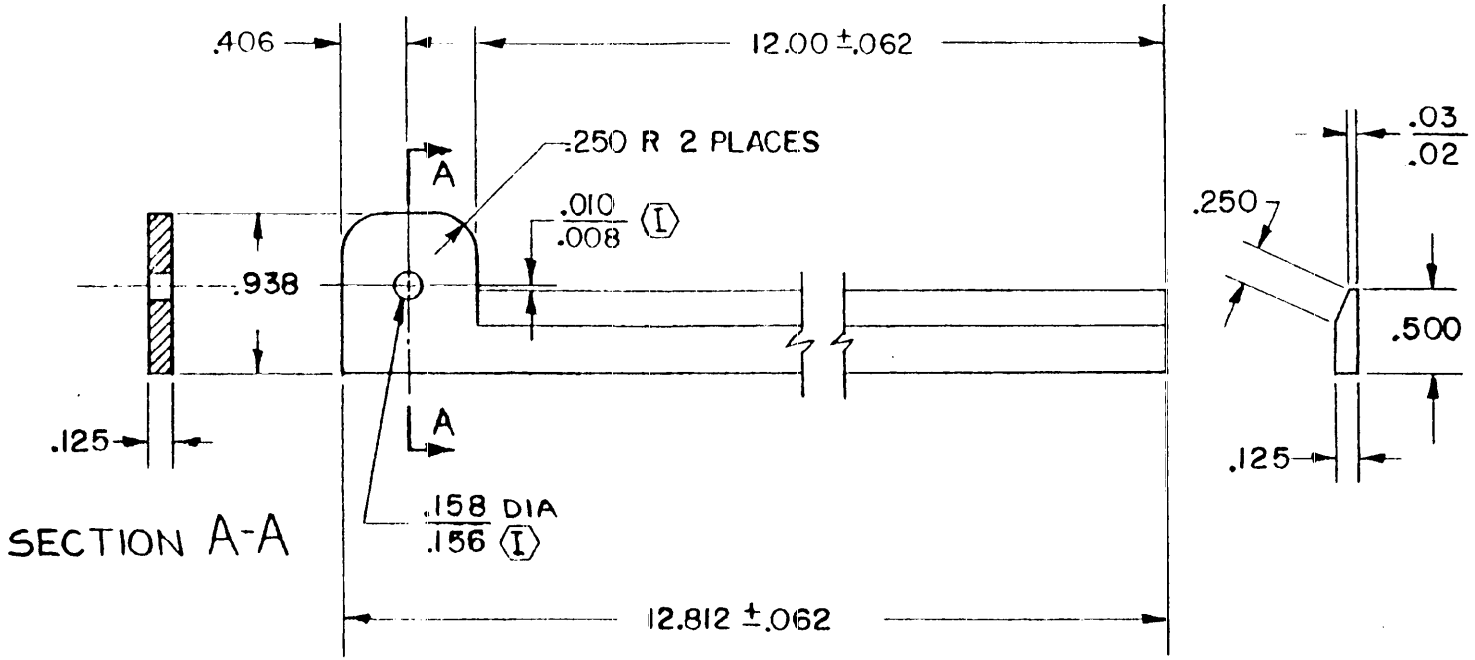
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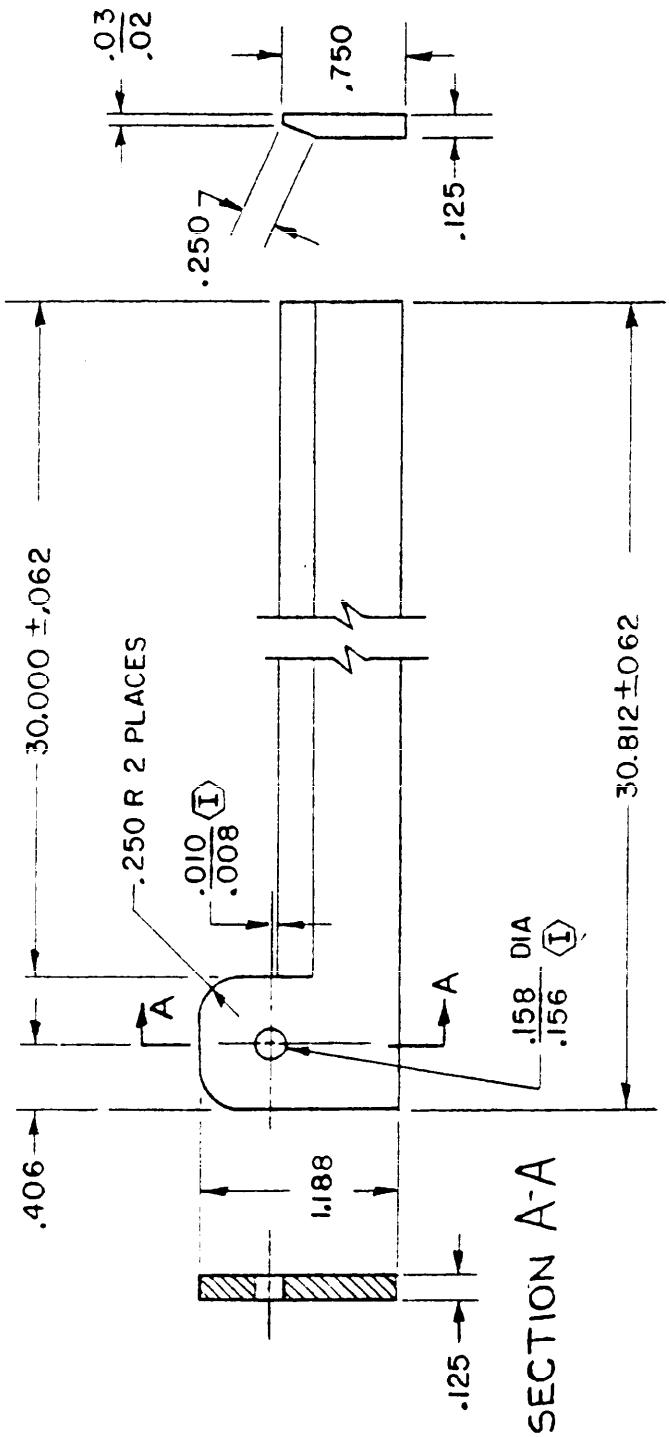
NOTES

1. SURFACE ROUGHNESS TO BE NOT MORE THAN 63 IN ACCORDANCE WITH ANSI B46.1.
2. Ⓘ INDICATES CRITICAL INSPECTION IS REQUIRED.
3. ALL DIMENSIONS ARE IN INCHES.
4. UNSPECIFIED TOLERANCE: ±.015.

FIGURE 1. 12 inch radial line straightedge.

X-171B

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FIGURE 2. 30 inch radial line straightedge.

X-172B

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